

Question 1: Faceting / Small Multiples

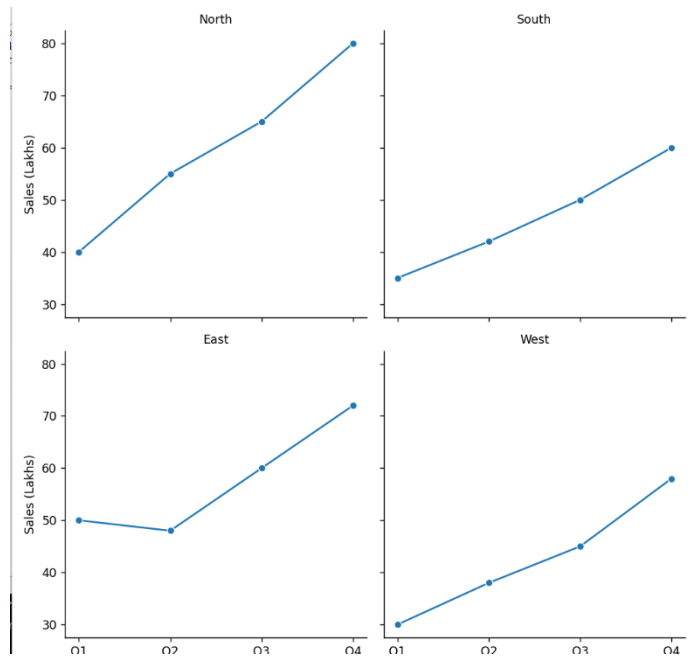
Code:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

data = {
    "Region": ["North","North","North","North",
               "South","South","South","South",
               "East","East","East","East",
               "West","West","West","West"],

    "Quarter": ["Q1","Q2","Q3","Q4"]*4,

    "Sales": [40,55,65,80,
              35,42,50,60,
              50,48,60,72,
              30,38,45,58]
}
```



```
df = pd.DataFrame(data)
```

```
g = sns.FacetGrid(df, col="Region", col_wrap=2, height=4)
```

```
g.map_dataframe(sns.lineplot, x="Quarter", y="Sales", marker="o")
```

```
g.set_axis_labels("Quarter", "Sales (Lakhs)")
```

```
g.set_titles("{col_name}")
```

```
plt.suptitle("Quarterly Sales Trend by Region", y=1.05)
```

```
plt.show()
```

Question 2: Legends and Annotations

Code:

```
import matplotlib.pyplot as plt
```

```
months = ["Jan","Feb","Mar","Apr","May","Jun"]
```

```
chennai = [25,27,31,34,37,35]
```

```
delhi = [14,18,24,30,34,36]
```

```
shimla = [5,8,13,18,22,24]
```

```
plt.figure(figsize=(8,5))
```

```
plt.plot(months, chennai, marker='o',  
label="Chennai")
```

```
plt.plot(months, delhi, marker='o',  
label="Delhi")
```

```
plt.plot(months, shimla, marker='o', label="Shimla")
```

```
plt.annotate("Highest",  
            xy=("May",37),  
            xytext=("Apr",39),  
            arrowprops=dict(facecolor='black'))
```

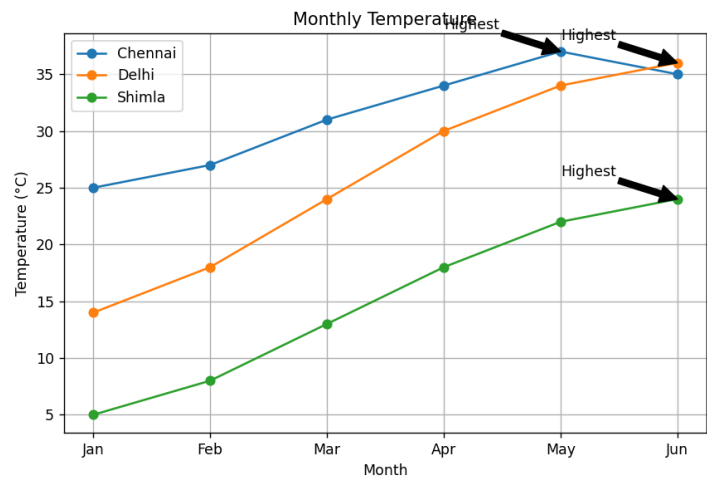
```
plt.annotate("Highest",  
            xy=("Jun",36),  
            xytext=("May",38),  
            arrowprops=dict(facecolor='black'))
```

```
plt.annotate("Highest",      xy=("Jun",24),  
            xytext=("May",26),  
            arrowprops=dict(facecolor='black'))
```

```
plt.xlabel("Month")
```

```
plt.ylabel("Temperature (°C)")
```

```
plt.title("Monthly Temperature")
```



```
plt.legend()
plt.grid(True)
```

```
plt.show()
```

Question 3: Mapping Data onto Aesthetics

Code;

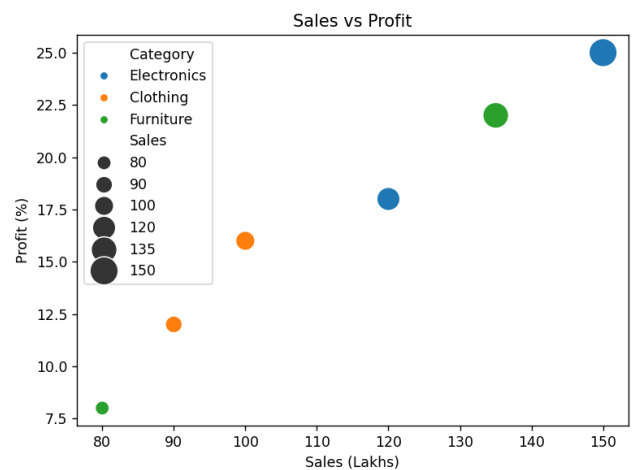
```
import pandas as pd
```

```
import seaborn as sns
```

```
import matplotlib.pyplot as plt
```

```
data = {
    "Product":["P1","P2","P3","P4","P5","P6"],
    "Sales":[120,90,150,80,135,100],
    "Profit":[18,12,25,8,22,16],
    "Category":["Electronics","Clothing","Electronics",
    "Furniture","Furniture","Clothing"]
}
```

```
df = pd.DataFrame(data)
```



```
plt.figure(figsize=(7,5))
```

```
sns.scatterplot(
    data=df,
    x="Sales",
    y="Profit",
    hue="Category",
    size="Sales",
    sizes=(100,400)
)
```

```
plt.title("Sales vs Profit")
```

```
plt.xlabel("Sales (Lakhs)")
```

```
plt.ylabel("Profit (%)")
```

```
plt.show()
```

Question 4: Nonlinear Axes

Code

```
library(ggplot2)
```

```
day <- c(1,2,3,4,5,6,7)
```

```
visitors <- c(20,35,70,150,320,650,1280)
```

```
df <- data.frame(day, visitors)
```

```
# Linear Scale
```

```
ggplot(df, aes(day, visitors)) +
```

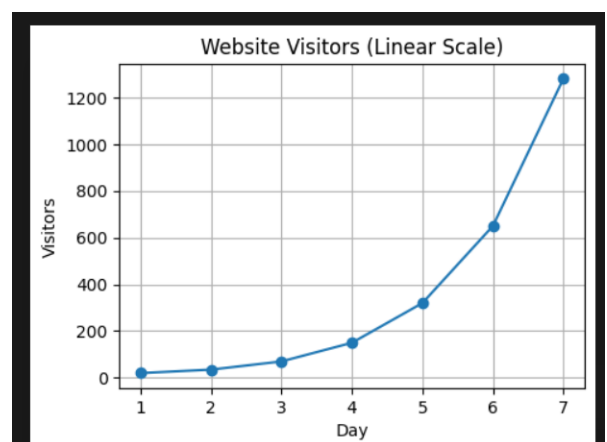
```
  geom_line(color="blue") +
```

```
  geom_point(size=3) +
```

```
  ggtitle("Website Visitors (Linear Scale)") +
```

```
  xlab("Day") +
```

```
  ylab("Visitors")
```



```
# Logarithmic Scale
```

```
ggplot(df, aes(day, visitors)) +
```

```
  geom_line(color="red") +
```

```
  geom_point(size=3) + scale_y_log10() +
```

```
  ggtitle("Website Visitors (Log Scale)") +
```

```
  xlab("Day") +
```

```
  ylab("Visitors")
```

Question 5: Cartesian and Polar Coordinates

Code:

```
import matplotlib.pyplot as plt
```

```
direction =
["N","NE","E","SE","S","SW","W","NW"]

frequency = [18,12,20,15,22,11,16,14]
```

```
plt.figure(figsize=(7,5))
```

```
plt.bar(direction, frequency)
```

```
plt.xlabel("Direction")
```

```
plt.ylabel("Frequency")
```

```
plt.title("Direction Frequency")
```

```
plt.show()
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
direction =
["N","NE","E","SE","S","SW","W","NW"]

frequency = [18,12,20,15,22,11,16,14]
```

```
angles = np.linspace(0, 2*np.pi, len(direction),
endpoint=False)
```

```
fig = plt.figure(figsize=(6,6))
```

```
ax = fig.add_subplot(111, polar=True)
```

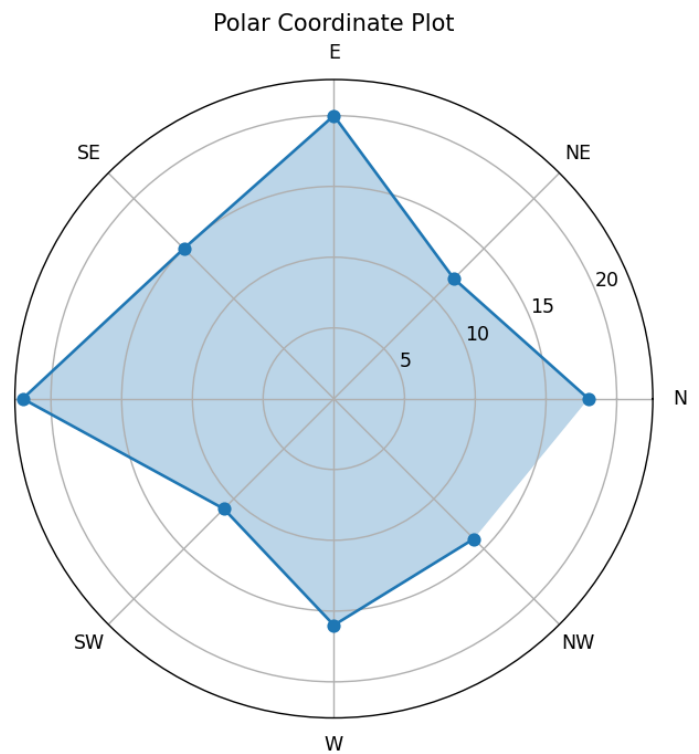
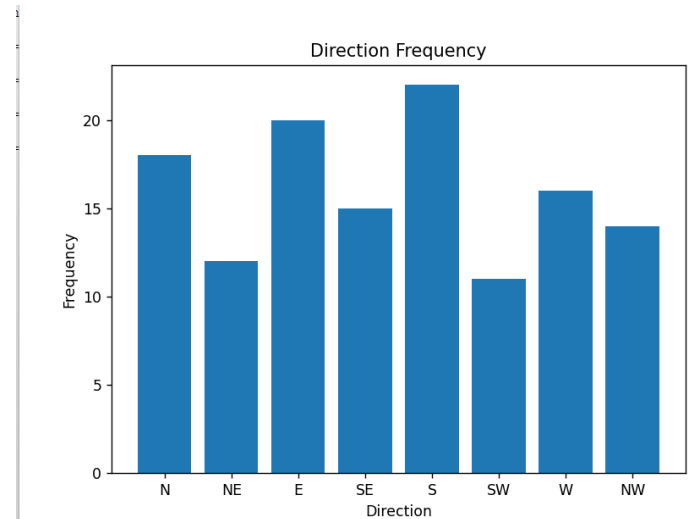
```
ax.plot(angles, frequency, marker='o')
```

```
ax.fill(angles, frequency, alpha=0.3)
```

```
ax.set_xticks(angles)
```

```
ax.set_xticklabels(direction)
```

```
plt.title("Polar Coordinate Plot")
```



```
plt.show()
```